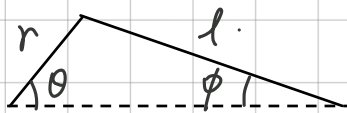


(1)



$$r \cdot \sin \theta = l \cdot \sin \phi \quad \theta = \omega t.$$

$$y(t) = r + l - r \cdot \cos \theta - l \cdot \cos \phi.$$

$$= r + l - r \cos \omega t - l \cdot \sqrt{1 - \sin^2 \theta} = r + l - r \cos \omega t - l \cdot \sqrt{1 - \frac{r^2}{l^2} \sin^2 \omega t}$$

$$\therefore \frac{r}{l} = 0.1 \ll 1$$

$$\therefore \sqrt{1 - \frac{r^2}{l^2} \sin^2 \omega t} \approx 1 - \frac{1}{2} \frac{r^2}{l^2} \sin^2 \omega t = 1 - \frac{1}{2} \frac{r^2}{l^2} \frac{1}{2} (1 - \cos 2\omega t).$$

$$\therefore y(t) = r + \frac{r^2}{4l} - r \cos \omega t - \frac{r^2}{4l} \cos 2\omega t = 0.1025 - 0.1 \cos(100t) - 0.0025 \cos(200t)$$

$$m\ddot{x} = k(y - x) + c(\dot{y} - \dot{x}) \quad \therefore m\ddot{x} + c\dot{x} + kx = c\dot{y} + ky$$

$$H(\Omega) = \frac{X(\Omega)}{Y(\Omega)} = \frac{k + i c \Omega}{k - m \Omega^2 + i c \Omega}$$

$$H(100) = \frac{100 + i 1000}{-9900 + i 1000} \approx 0.0001 - i 0.1. \quad \therefore |H(100)| \approx 0.1$$

$$\angle H(100) \approx -89.94^\circ$$

$$H(200) = \frac{100 + i 2000}{100 - 40000 + i 2000} \approx 6.27 \times 10^{-6} - i 0.050125 \quad |H(200)| \approx 0.050125$$

$$\angle H(200) \approx -89.99^\circ$$

$$\therefore x(t) \approx 0.0105 \cos(100t - 89.94^\circ) + 1.253 \times 10^{-4} \cos(200t - 89.99^\circ)$$

$$(2) \quad \omega_0 = \frac{2\pi}{T}$$

$$y(t) = \frac{Y}{T} t \quad (0 \leq t \leq T)$$

$$\Rightarrow y(t) = \frac{Y}{2} - \frac{Y}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin(n\omega_0 t)$$

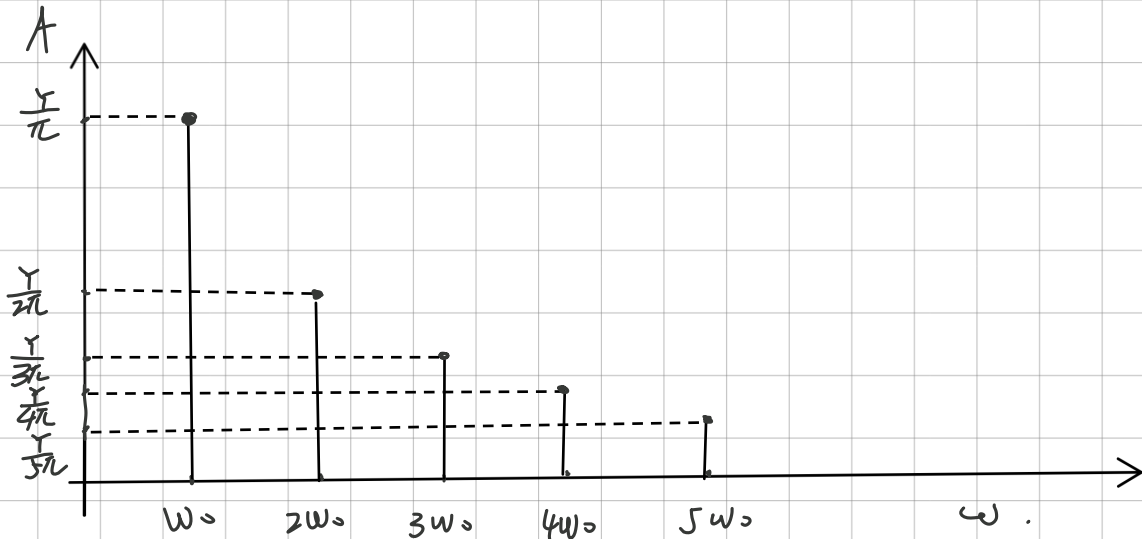
$$m\ddot{x} + c\dot{x} + kx = c\dot{y} + ky$$

$$G(s) = \frac{X(s)}{Y(s)} = \frac{cs + k}{ms^2 + cs + k} \quad H(\Omega) = G(i\Omega) = \frac{k + ic\Omega}{k - m\Omega^2 + ic\Omega}$$

$$Y_n = \frac{Y}{n\pi} \quad X_n = \frac{Y}{n\pi} |H(n\omega_0)| \quad \phi_n = \angle H(n\omega_0)$$

$$\therefore x(t) \approx \frac{Y}{2} - \sum_{n=1}^{\infty} \frac{Y}{n\pi} |H(n\omega_0)| \sin[n\omega_0 t + \phi_n]$$

$$\text{解: } |H(n\omega_0)| = \left| \frac{k + ic \cdot n\omega_0}{k - m(n\omega_0)^2 + icn\omega_0} \right| \quad \phi_n = \angle H(n\omega_0)$$



13)

$$m\ddot{x} = F(t) - kx \quad \therefore m\ddot{x} + kx = F(t).$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

$$x(0) = 0 \quad \dot{x}(0) = 0$$

$$F(t) = \begin{cases} \frac{F_0}{t_0} t & 0 \leq t \leq t_0 \\ F_0 & t_0 \leq t \leq 5t_0 \\ 0 & 5t_0 \leq t \end{cases}$$

$$\Rightarrow F(t) = \frac{F_0}{t_0} t u(t) - \frac{F_0}{t_0} (t - t_0) u(t - t_0) - F_0 u(t - 5t_0)$$

$$m\ddot{x} + kx = F(t) \Rightarrow (ms^2 + k)X(s) = F(s).$$

$$\therefore X(s) = \frac{F(s)}{ms^2 + k} = \frac{F(s)}{m(s^2 + \omega_n^2)}$$

$$F(s) = \frac{F_0}{t_0 s^2} - \frac{F_0}{t_0} \frac{e^{-st_0}}{s^2} - \frac{F_0 e^{-5st_0}}{s}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2(s^2 + \omega_n^2)} \right\} = \frac{1}{\omega_n^2} \left(t - \frac{\sin \omega_n t}{\omega_n} \right)$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{s(s^2 + \omega_n^2)} \right\} = \frac{1}{\omega_n^2} (1 - \cos \omega_n t)$$

$$\therefore x(t) = \begin{cases} 0 & t < 0 \\ \frac{F_0}{k t_0} \left(t - \frac{\sin \omega_n t}{\omega_n} \right) & 0 \leq t < t_0 \\ \frac{F_0}{k} \left[1 - \frac{\sin \omega_n t - \sin \omega_n (t - t_0)}{\omega_n t_0} \right] & t_0 \leq t < 5t_0 \\ \frac{F_0}{k} \left[\cos \omega_n (t - 5t_0) - \frac{\sin \omega_n t - \sin \omega_n (t - t_0)}{\omega_n t_0} \right] & t \geq 5t_0 \end{cases}$$

